1. Project Title

Car Rental Management System

A comprehensive database solution for managing car rental operations including branch management, fleet tracking, customer registration, rental transactions, and payment processing.

2. Group Members' Names

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3. Project Description

3.1 Overview

The **Car Rental Management System** is a desktop application developed in C++ with MySQL backend that automates the operations of a car rental business. The system enables efficient management of multiple branches, vehicle fleets, customer registrations, rental transactions, and payment processing.

3.2 Key Features

Feature	Description
Branch Management	Add, view, and manage multiple rental locations
Car Management	Add vehicles, track availability, manage fleet
Customer Management	Register customers, store contact and license information
Rental Operations	Rent cars, process returns, calculate charges

Feature	Description
Overdue Tracking	Automatically identify and calculate fines for late returns
Payment Processing	Record payments via Cash, Card, or Online methods
Reporting	Revenue reports, rental history, overdue reports

3.3 Scope

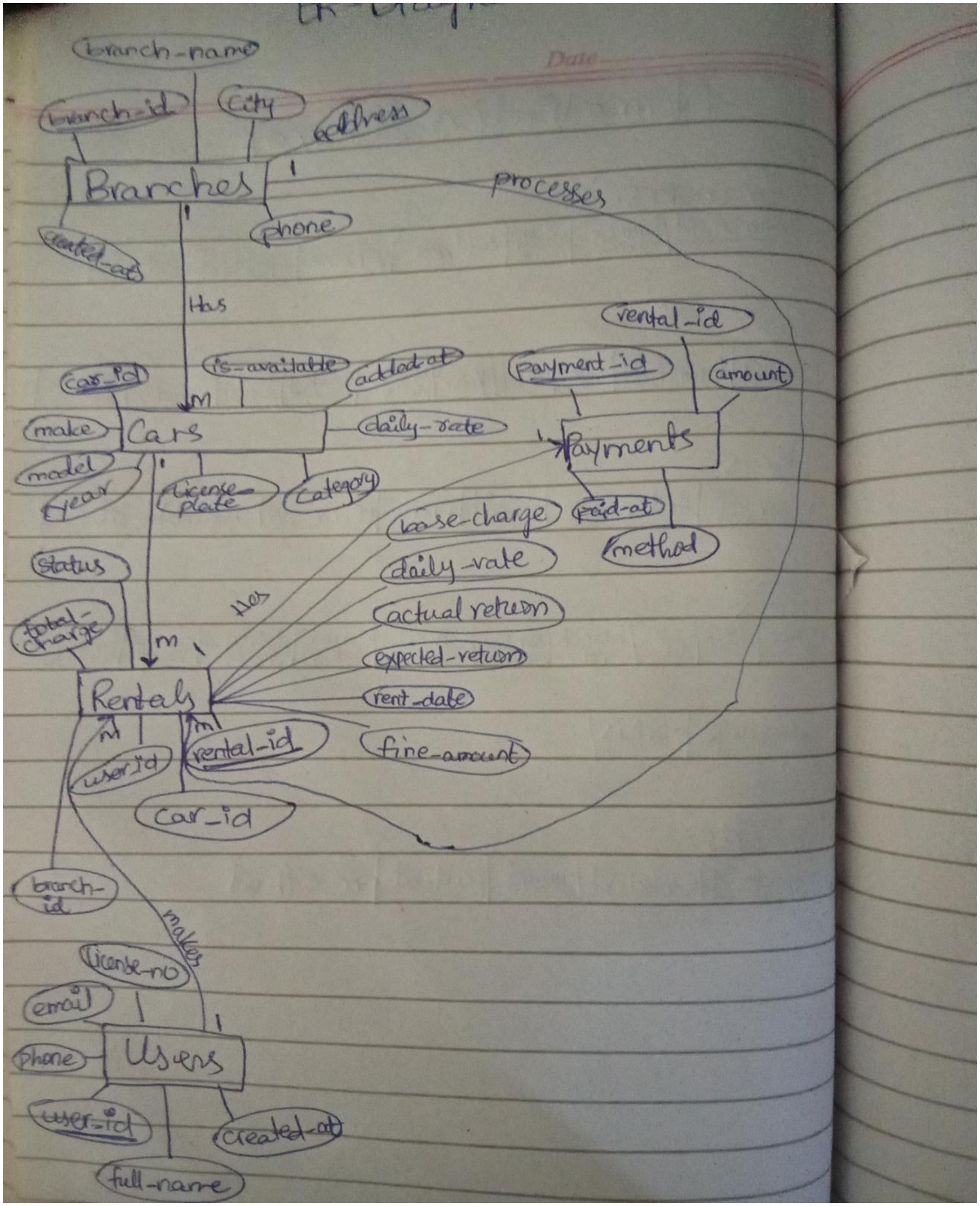
The system handles:

- Multiple branches across different cities
- Different car categories (Economy, Compact, SUV, Luxury, Van)
- Customer data management with unique license numbers
- Rental transactions with date-based calculations
- Fine calculation for overdue returns (20% per day penalty)
- Revenue tracking by branch

4.Project requirements

- Windows OS (64-bit)
- Visual Studio 2022
- MySQL Server 8.0
- MySQL C API
- 4 GB RAM minimum
- 500 MB storage

5.ER Diagram



branch-name
branch-id
City
address
Branches
phone
created-at
processes
Has
rental-id
payment-id
amount
car-id
is-available
added-at
make
Cars
daily-rate
Payments
model
year
license-plate
category
base-charge
paid-at
method
daily-rate
status
actual return
total-charge
Has
expected-return
Rentals
rent-date
fine-amount
user-id
rental-id
car-id
branch-id
makes
license-no
email
Users
phone
user-id
created-at
full-name

6. Normalized Schema

Normalization Schemas

Branches:

branch-id	branch-name	city	address	phone	created-at

Cars:

car-id	branch-id	make	model	year	license-plate	category	daily-rate	is-available	added-at

Users:

user-id	full-name	email	phone	license-no	created-at

Rentals:

rental-id	user-id	car-id	branch-id	rent-date	expected-return	actual-return	daily-rate	base-charge	fine-amount

total-charge	status

Payments:

payment-id	rental-id	amount	paid-at	method

Functional Dependencies:

- Branches:

branch_id → branch_name, city, address, phone, created_at.

- Cars:

car_id → make, model, year, license_plate, category, daily_rate, is_available, added_at, branch_id.

- Users:

user_id → full_name, email, phone, license_no, created_at

- Rentals:

rental_id → user_id, car_id, branch_id, rent_date, expected_return, actual_return, daily_rate, base_charge, fine_amount, total_charge, status.

- Payments:

payment_id → rental_id, amount, paid_at, method.